

A partial balance equation based solution

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The Markov chain state transition diagram is as follows.

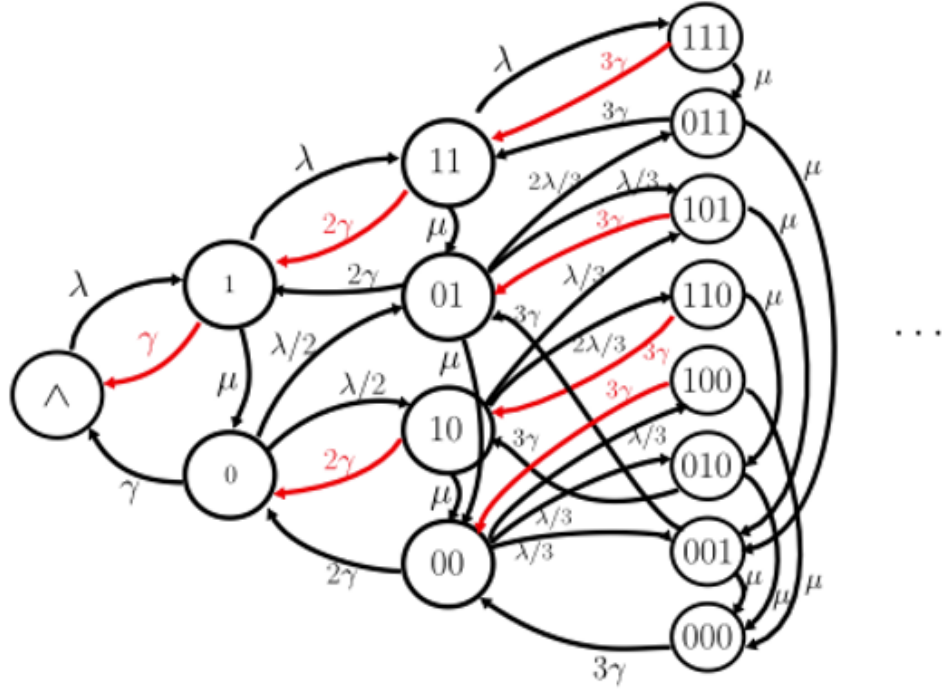


Figure 1: Markov chain

Known equations

$$P_{\wedge} = e^{-\lambda/\gamma}$$

$$\lambda P_{\wedge} = \gamma(P_0 + P_1)$$

$$\lambda(P_0 + P_1) = 2\gamma(P_{00} + P_{01} + P_{10} + P_{11})$$

Global balance equation at P_1

$$\lambda P_{\wedge} + 2\gamma P_{11} + 2\gamma P_{01} = \lambda P_1 + \mu P_1 + \gamma P_1$$

Verified Partial Balance Equations at P_1 :

$$\lambda P_{\wedge} + \gamma P_{01} = (\mu + \gamma)P_1 \tag{1.1}$$

```
{'Eqn 1.1': {'R2': 0.9991109292735545,
'MSE': 0.028585677850327677,
'NMSE': 0.0007537520117123942},
'Eqn 1.2': {'R2': 0.9957046749896326,
'MSE': 0.028585671273886424,
'NMSE': 0.0007537518383036736}}
```

Figure 2: R2 score for partial balance equations at P1

$$2\gamma P_{11} + \gamma P_{01} = \lambda P_1 \quad (1.2)$$

Global balance equation at P_0

$$\mu P_1 + 2\gamma(P_{10} + P_{00}) = \lambda P_0 + \gamma P_0$$

Verified Partial Balance Equations at P_0 :

$$\mu P_1 + \gamma P_{00} = \left(\frac{\lambda}{2} + \gamma\right) P_0 \quad (2.1)$$

$$2\gamma P_{10} + \gamma P_{00} = \frac{\lambda}{2} P_0 \quad (2.2)$$

```
{'Eqn 2.1': {'R2': 0.9975655075922147,
'MSE': 0.028585678945676974,
'NMSE': 0.0010313676842910724},
'Eqn 2.2': {'R2': 0.9857596741168904,
'MSE': 0.02858567442528982,
'NMSE': 0.001031367521196075}}
```

Figure 3: R2 score for partial balance equations at P0

Further Simplifying

Substituting the value of $\frac{\lambda}{2} P_0$ from Equation (2.2) into RHS of Equation (2.1) we get

$$\begin{aligned}
\mu P_1 + \gamma P_{00} &= \left(\frac{\lambda}{2} + \gamma\right) P_0 \\
&= \frac{\lambda}{2} P_0 + \gamma P_0 \\
&= 2\gamma P_{10} + \gamma P_{00} + \gamma P_0 \\
\Rightarrow \mu P_1 + \gamma P_{00} &= 2\gamma P_{10} + \gamma P_{00} + \gamma P_0 \\
\Rightarrow \mu P_1 &= 2\gamma P_{10} + \gamma P_0 \\
&= 2\gamma P_{10} + \lambda P_\wedge - \gamma P_1 \quad (\text{Since } \gamma(P_0 + P_1) = \lambda P_\wedge) \\
\Rightarrow \mu P_1 + \gamma P_1 &= 2\gamma P_{10} + \lambda P_\wedge \\
\Rightarrow \gamma P_{01} + \lambda P_\wedge &= 2\gamma P_{10} + \lambda P_\wedge \quad \text{From Eqn. (1.1)} \\
\Rightarrow \gamma P_{01} &= 2\gamma P_{10}
\end{aligned} \quad (3.1)$$

Now substituting Eqn (3.1) into Eqn (1.2) we get

$$\begin{aligned} 2\gamma P_{11} + \gamma P_{01} &= \lambda P_1 \\ 2\gamma P_{11} + 2\gamma P_{10} &= \lambda P_1 \end{aligned} \quad (3.2)$$

and since we know that $2\gamma(P_{11} + P_{10} + P_{01} + P_{00}) = \lambda(P_1 + P_0)$, we can also say that

$$2\gamma P_{00} + 2\gamma P_{01} = \lambda P_0 \quad (3.3)$$

Verifying these from simulations data and from probabilities obtained from matrix inversion for finite buffer length of 20:

```
{ 'Eqn 3.2': { 'R2': 0.9999999999999929,
  'MSE': 4.9411873237704546e-14,
  'NMSE': 4.214685642379657e-15},
  'Eqn 3.3': { 'R2': 0.9999999999999931,
  'MSE': 4.990338867601881e-14,
  'NMSE': 4.25661044557231e-15},
  'Eqn 4.1': { 'R2': 0.99999999999999488,
  'MSE': 1.1226864390377549e-13,
  'NMSE': 9.576180997518051e-15},
  'Eqn 4.2': { 'R2': 0.99999999999999163,
  'MSE': 1.1068038598685238e-13,
  'NMSE': 9.440707326915665e-15}}
```

Generalising the above to third column we have

$$\lambda P_{00} = 3\gamma(P_{001} + P_{000}) \quad (4.1)$$

$$\lambda P_{11} = 3\gamma(P_{111} + P_{110}) \quad (4.2)$$

where R2 scores of these are shown in the above figure.

In general

$$\lambda P_x = |x|\gamma(P_{x0} + P_{x1})$$

where 'x' can be any state and |x| is the length of string 'x'

Solving P0 and P1

Currently we have the following set of independent equations:

$$\lambda P_\Lambda + \gamma P_{01} = (\mu + \gamma)P_1 \quad (5.1)$$

$$2\gamma P_{11} + \gamma P_{01} = \lambda P_1 \quad (5.2)$$

$$\mu P_1 + \gamma P_{00} = \left(\frac{\lambda}{2} + \gamma\right)P_0 \quad (5.3)$$

$$2\gamma P_{10} + \gamma P_{00} = \frac{\lambda}{2}P_0 \quad (5.4)$$

$$\lambda(P_0 + P_1) = 2\gamma(P_{00} + P_{01} + P_{10} + P_{11}) \quad (5.5)$$

Combining (5.1) and (5.2) we get GBE at P1. Combining (5.3) and (5.4) we get GBE at P0. Combining (5.1), (5.2), (5.3), (5.4) we get $\lambda P_\Lambda = P_0 + P_1$

So, we have 5 independent equations (5.1) to (5.5) and 6 unknowns ($P_0, P_1, P_{00}, P_{01}, P_{10}, P_{11}$), so the set of equations is still inconsistent.